MSc. Project Report: Survey on Equations over Finite Fields

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Soham Chatterjee

soham.chatterjee@tifr.res.in
Website: sohamch08.github.io

Abstract

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Preface

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Acknowledgements

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List of Notations

The following table lists the notation used throughout this report, organized by topic.

Symbol	Meaning
General	
$\mathbb{N}$	Set of all natural numbers i.e., positive integers
$\mathbb{Z}$	Set of all integers
$\mathbb{Z}_0$	Set of all non-negative integers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
$[n]$	The set $\{1, 2, \ldots, n\}$ for $n \in \mathbb{N}$
i	Imaginary unit: $i^2 = -1$
S^1	$\{z \in \mathbb{C} : \lvert z\rvert = 1\}$, the multiplicative group of unit complex numbers
ζ_m	A primitive m-th root of unity
$[x]$	Floor function: largest integer $\leq x$; also written $\lfloor x \rfloor$
Finite Fields	
$\mathbb{F}_q$	Finite field with q elements
$\mathbb{F}_q^*$	Multiplicative group of nonzero elements of $\mathbb{F}_q$
$\mathbb{F}_p$	Prime field with p elements, $p = \mathrm{char}(\mathbb{F}_q)$
$\mathbb{F}_q^{*\,(d)}$	Subgroup of d-th powers in $\mathbb{F}_q^*$
E, F, L, K	Generic notation for finite fields or finite extensions of $\mathbb{F}_q$
$[E : \mathbb{F}_q]$	Degree of the extension $E/\mathbb{F}_q$
$\mathbb{F}_q[X_1, \ldots, X_n]$	Polynomial ring in n variables over $\mathbb{F}_q$
$\mathbb{F}_q^{\leq d}[X_1, \ldots, X_n]$	Polynomials in $\mathbb{F}_q[X_1, \ldots, X_n]$ of total degree at most d
Characters of Finite Abelian Groups	
G	A finite Abelian group with identity element 1_G
Z_k	Cyclic multiplicative group of order k
χ_0	Trivial character of G: $\chi_0(g) = 1$ for all $g \in G$
$\overline{\chi}$	Conjugate (and inverse) character of χ: $\overline{\chi}(g) = \overline{\chi(g)}$
$\mathrm{Hom}(A, B)$	Set of all group homomorphisms from A to B
$\widehat{G} = \mathrm{Hom}(G, S^1)$	Dual group of G; the group of all characters of G
$\mathrm{Ann}(\chi)$	Annihilator of character χ: $\{g \in G : \chi(g) = 1\}$, a subgroup of G
$\mathrm{Ann}(H)$	Annihilator of subgroup $H \leq G$: $\{\chi \in \widehat{G} : \chi(h) = 1\ \forall h \in H\}$

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Symbol	Meaning
ev_g	Evaluation character of $\widehat{G}$ defined by $\mathrm{ev}_g(\chi) = \chi(g)$
Maps on Finite Fields	
$\mathrm{Tr}\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute trace function
$N\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute norm function
$\mathrm{Tr}_{E/\mathbb{F}_q}$	Trace function from E to $\mathbb{F}_q$
$\mathrm{N}_{E/\mathbb{F}_q}$	Norm function from E to $\mathbb{F}_q$
$\sigma\colon \mathbb{F}_q \to \mathbb{F}_q$	Frobenius automorphism: $\sigma(\alpha) = \alpha^p$
Additive Characters of $\mathbb{F}_q$	
χ, τ	Generic notation for additive characters
χ_1	Canonical additive character of $\mathbb{F}_q$: $\chi_1(\alpha) = e^{2\pi i\,\mathrm{Tr}(\alpha)/p}$
χ_β	Additive character defined by $\chi_\beta(\alpha) = \chi_1(\beta\alpha)$, for $\beta \in \mathbb{F}_q$
$\boldsymbol{\mathcal{X}}_q$	Set of all additive characters of $\mathbb{F}_q$
$\chi^{(r)}$	Additive character of $\mathbb{F}_q$ lifted to E via $\chi^{(r)} = \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$, $[E : \mathbb{F}_q] = r$
Multiplicative Characters of $\mathbb{F}_q$	
ψ, λ	Generic notation for multiplicative characters
ψ_0	Trivial multiplicative character of $\mathbb{F}_q$
η	Quadratic character of $\mathbb{F}_q$ (for odd q): $\eta(\alpha) = \pm 1$ according as α is a square
$\boldsymbol{\mathcal{M}}_q$	Set of all multiplicative characters of $\mathbb{F}_q^*$
$\boldsymbol{\mathcal{M}}_q^{(e)}$	Set of multiplicative characters of $\mathbb{F}_q$ of exponent e
$\mathrm{ord}(\psi)$	Order of ψ: smallest positive integer d with $\psi^d = \psi_0$
$\left(\frac{\alpha}{q}\right)$	Legendre symbol; equals $\eta(\alpha)$ for $\alpha \in \mathbb{F}_q$
$\psi^{(r)}$	Multiplicative character of $\mathbb{F}_q$ lifted to E via $\psi^{(r)} = \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$, $[E\colon \mathbb{F}_q] = r$
Gaussian and Jacobi Sums	
$G(\psi, \chi)$	Gaussian sum: $\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha)\,\chi(\alpha)$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$
$J_\alpha(\lambda_1, \ldots, \lambda_k)$	Generalized Jacobi sum: $\displaystyle\sum_{\substack{\alpha_1,\ldots,\alpha_k \in \mathbb{F}_q \\ \alpha_1+\cdots+\alpha_k=\alpha}} \lambda_1(\alpha_1)\cdots\lambda_k(\alpha_k)$
$J(\lambda_1, \ldots, \lambda_k)$	Jacobi sum; special case $\alpha = 1$ of J_α
$J_0(\lambda_1, \ldots, \lambda_k)$	Jacobi sum at zero; special case $\alpha = 0$ of J_α
Kloosterman Sums	
$K(\chi; a, b)$	Kloosterman sum: $\sum_{\gamma \in \mathbb{F}_q^*} \chi(a\gamma + b\gamma^{-1})$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $a, b \in \mathbb{F}_q$
$K(\chi^{(r)}; a, b)$	Kloosterman sum lifted to the degree-r extension of $\mathbb{F}_q$
w_1, w_2	Reciprocal roots of the L-function associated to a Kloosterman sum
Salié and Jacobsthal Sums	
$S(\chi, \tau)$	Salié sum: $\sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\,\chi(\gamma)\,\tau(\gamma^{-1})$, for $\chi, \tau \in \boldsymbol{\mathcal{X}}_q$, q odd
$H_n(\alpha)$	Jacobsthal sum: $\sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^{n+1} + \alpha\gamma)$, for $\alpha \in \mathbb{F}_q^*$

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Symbol	Meaning
$I_n(\alpha)$	Associated Jacobsthal sum: $\sum_{\gamma\in\mathbb{F}_q} \eta(\gamma^n+\alpha)$, for $\alpha \in \mathbb{F}_q^*$
Classical Riemann Zeta Function	
$\zeta(s)$	Riemann zeta function: $\sum_{n=1}^{\infty} n^{-s}$, for $\Re(s) > 1$
$\xi(s)$	Completed zeta function: $\frac{1}{2}s(s-1)\,\Gamma(s/2)\,\pi^{-s/2}\,\zeta(s)$
$\Gamma(s)$	Gamma function: $\int_0^{\infty} e^{-t}t^{s-1}\,dt$
$\Lambda(n)$	Von Mangoldt function: $\log p$ if $n = p^k$, else 0
ρ	Non-trivial zero of $\xi(s)$
Dirichlet L-functions over $\mathbb{F}_q[X]$	
Φ	Set of all monic polynomials in $\mathbb{F}_q[X]$
Φ_d	Monic polynomials in $\mathbb{F}_q[X]$ of degree exactly d
$\mathrm{Irr}(\Phi)$	Set of irreducible monic polynomials in Φ
$\mathfrak{n}(f) = q^{\deg(f)}$	Norm of the polynomial $f \in \mathbb{F}_q[X]$
G	Group of rational functions $r = h_1/h_2$ in $\mathbb{F}_q(X)$ with $h_1, h_2 \in \Phi$ monic
$\overline{G}$	Subgroup of G on which a given character is defined; extended to G by zero
$L(s,\chi)$	Dirichlet L-function: $\sum_{h\in\Phi} \chi(h)\,\mathfrak{n}(h)^{-s}$, for a multiplicative character χ
$\overline{L}(z,\chi)$	Power series satisfying $\overline{L}(z,\chi) = \sum_{h\in\Phi} \chi(h)\,z^{\deg(h)}$ and $\overline{L}(q^{-s},\chi) = L(s,\chi)$
L_k	k-th log-derivative coefficient: $\displaystyle\sum_{\substack{h\in\mathrm{Irr}(\Phi)\\ \deg(h)\mid k}} \deg(h)\,\chi(h)^{k/\deg(h)}$
Equations over $\mathbb{F}_q$ *— Solution Sets*	
$\mathcal{V}(f)$	Solution set of $f(X_1,\ldots,X_n) = 0$ in $\mathbb{F}_q^n$
$Z(f)$	Number of solutions: $\lvert\mathcal{V}(f)\rvert$
$\overline{\mathcal{V}}(f)$	Set of *non-trivial* solutions (excluding $(0,\ldots,0)$ if it's a zero) of $f = 0$
$\overline{Z}(f)$	Number of non-trivial solutions: $\lvert\overline{\mathcal{V}}(f)\rvert$
$\mathcal{V}(f_1,\ldots,f_k)$	Common solution set: $\mathcal{V}(f_1)\cap\cdots\cap\mathcal{V}(f_k)$
$Z(f_1,\ldots,f_k)$	Count of common solutions: $\lvert\mathcal{V}(f_1,\ldots,f_k)\rvert$
$Z_E(P=0)$	Number of solutions of $P(X_1,\ldots,X_n) = 0$ in E where $P \in \mathbb{F}_q[X_1,\ldots,X_n]$, $[E\colon \mathbb{F}_q] > 1$
Ω_d	Space $\mathbb{F}_q^{\leq d}[X_1,\ldots,X_n]$ of polynomials of total degree at most d
$\omega(d)$	Set of n-tuples $(i_1,\ldots,i_n) \in \mathbb{Z}_0^n$ with $i_1+\cdots+i_n \leq d$
Quadratic Forms	
C_f	Coefficient matrix of a quadratic form f: $(C_f)_{ij} = a_{ij}$ where $f = X^{\top}C_fX$
$\det(f)$	Determinant of the coefficient matrix C_f
$\vartheta : \mathbb{F}_q \to \mathbb{Z}$	Auxiliary function: $\vartheta(0) = q-1$ and $\vartheta(\alpha) = -1$ for $\alpha \neq 0$
Diagonal Equations	
$d_i = \gcd(k_i, q-1)$	Exponent reduced modulo the order of $\mathbb{F}_q^*$, for $a_iX_i^{k_i}$ in a diagonal equation
$M(d_1,\ldots,d_n)$	$\#\left\{(t_1,\ldots,t_n) \in \prod_i[d_i-1] : \frac{t_1}{d_1}+\cdots+\frac{t_n}{d_n} \in \mathbb{Z}\right\}$
$D = \mathrm{lcm}(d_1,\ldots,d_n)$	LCM of exponents; appears in the closed-form formula for $M(d_1,\ldots,d_n)$
Character Sums with Polynomials (Weil Sums)	

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Symbol	**Meaning**
$W(\chi; f)$	Additive Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \chi(f(\alpha))$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi; f)$	Multiplicative Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi, \chi; f, g)$	Mixed Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))\, \chi(g(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$, $f, g \in \mathbb{F}_q[X]$
Lifted Weil Sums (over Extension Fields)	
$E = \mathbb{F}_{q^k}$	Degree-k extension of $\mathbb{F}_q$; $[E : \mathbb{F}_q] = k$
$W^{(k)}(\psi^{(k)}; f)$	Lifted multiplicative Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))$
$W^{(k)}(\chi^{(k)}; g)$	Lifted additive Weil sum: $\sum_{\alpha \in E} \chi^{(k)}(g(\alpha))$
$W^{(k)}(\psi^{(k)}, \chi^{(k)}; f, g)$	Lifted mixed Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))\, \chi^{(k)}(g(\alpha))$
Special L-functions (Weil chapter)	
$\overline{G}$ (Weil context)	The specific subgroup $\{r \in G : h_1(\gamma_i) \cdot h_2(\gamma_i) \neq 0\ \forall i\}$ for roots γ_i of a fixed f
$\varsigma : \overline{G} \to \mathbb{F}_q$	Multiplicative map $\varsigma(r) = \prod_i f(\alpha_i) \cdot \prod_j f(\beta_j)^{-1}$ for $r = \prod(X - \alpha_i)/\prod(X - \beta_j)$
$\varrho : G \to \mathbb{F}_q$	Additive map $\varrho(r) = \sum_i g(\alpha_i) - \sum_j g(\beta_j)$ for a fixed polynomial g
$\xi : G \to \mathbb{C}$	Combined character: $\xi(r) = (\psi \circ \varsigma)(r) \cdot (\chi \circ \varrho)(r)$
$L(s, \xi)$	L-function on ξ: $\sum_{h \in \Phi} \xi(h)\, \mathfrak{n}(h)^{-s}$
$\overline{L}(z, \xi)$	Power series: $\sum_{h \in \Phi} \xi(h)\, z^{\deg(h)}$, related by $\overline{L}(q^{-s}, \xi) = L(s, \xi)$
$L^{(k)}(s, \xi^{(k)})$	L-function on the lifted character $\xi^{(k)}$ over $\mathbb{F}_{q^k}$
$\overline{L}^{(k)}(z, \xi^{(k)})$	Power series form of the lifted L-function
$\Phi^{(k)}$	Set of monic polynomials over $E = \mathbb{F}_{q^k}$
$\mathfrak{n}^{(k)}(h) = q^{k \cdot \deg(h)}$	Norm of $h \in \Phi^{(k)}$ over E; analog of $\mathfrak{n}(h) = q^{\deg(h)}$ for $\mathbb{F}_q$
Absolute Varieties	
$\mathcal{A}$	Variety defined over $\mathbb{F}_q$
$Z_k(\mathcal{A})$	Number of $\mathbb{F}_{q^k}$-rational points of $\mathcal{A}$: $\lvert\{(\alpha_1, \ldots, \alpha_n) \in \mathcal{A} : \alpha_i \in \mathbb{F}_{q^k}\}\rvert$

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Chapter 1

Finite Fields

Fields are the most fundamental algebraic structures in which one can add, subtract, multiply, and divide (by nonzero elements) without leaving the set. Commonly known examples of fields are set of rationals $\mathbb{Q}$, or real numbers $\mathbb{R}$, or complex numbers $\mathbb{C}$. But the set of integers $\mathbb{Z}$ form a ring but not a field, since most elements have no multiplicative inverse. A *finite field* is a field that contains only finitely many elements. We assume the reader is familiar with basic ring theory; for a thorough treatment see [Art11]. Before proceeding, we recall the definitions of rings and fields.

Definition 1.0.1: Rings and Fields

(i) A ring is called a ring with identity *if the ring has a multiplicative identity—that is, if there is an element e such that $ae = ea = a$ for all $a \in R$.*

(ii) A ring is called commutative *if $\cdot$ is commutative.*

(iii) A ring is called an integral domain *if it is a commutative ring with identity $e \neq 0$ in which $ab = 0$ implies $a = 0$ or $b = 0$.*

(iv) A ring is called a division ring *(or* skew field*) if the nonzero elements of R form a group under $\cdot$.*

(v) A commutative division ring is called a field.

For any ring R with identity, the *characteristic*, $\mathrm{char}(R)$ is the smallest positive integer n such that $n \cdot 1 = 0$, or 0 if no such n exists. One can show that the characteristic of an integral domain is always either 0 or a prime; see [Art11] for a proof. Since every field is an integral domain, the characteristic of a field is always 0 or a prime p.

§ 1.1 Field Extensions

We build up the theory of field extensions following Morandi [Mor96].

The study of field theory is primarily the study of field extensions. The classical problems of ruler-and-compass constructions and the solvability of polynomial equations by radicals were settled by analyzing appropriate field extensions. The central idea is to associate a group—the *Galois group*—to a field extension, converting field-theoretic questions into group-theoretic ones. Since the Galois group of a finite extension is finite, the rich structure theory of finite groups becomes available. In this section we build the foundational vocabulary: degrees, algebraic elements, minimal polynomials, and the key structural results about simple and algebraic extensions.

1.1.1 Degrees and Generated Subfields

Let $F \subseteq K$ are fields. Then K is called a *field extension* of F. We denote it by K/F. Here F is called the *base field*. We can think of K as a vector space over F where the addition of two vectors follows the addition operation of K and the multiplication of any $\alpha \in K$ and $c \in F$ follows the multiplication operation of K, $c \cdot \alpha = c\alpha$. Then we denote the dimension of K over F by

$$[K\colon F] \coloneqq \dim_F(K)$$

This dimension is called the *degree of field extension* of F or simply *degree of* $[K\colon F]$. If $[K\colon F] < \infty$ we call it by a special name, *finite extension*. Finite extensions of finite fields are of very important to us and this report we will work on finite fields and their finite extensions. Let $\alpha \in K$. Then we denote $F(\alpha)$ to be the field generated by F and α following the multiplication and addition operations of K. We will define what it means to be the field generated by F and α in Definition 1.1.1. It is very easy to sea $F(\alpha)$ is also a field, and it contains F but also contained in K.

Let F be any field and suppose X be a formal variable. Then $F[X]$ denotes the polynomial ring over F. We define the rational function field $F(X)$ to be the *fraction ring* of $F[X]$ which consists of all quotients f/g where $f, g \in F[X]$ with $g(X) \neq 0$. Similarly, we can define the polynomial ring $F[X_1, \ldots, X_n]$ over multiple formal variables and their fraction field $F(X_1, \ldots, X_n)$.

Look at the following example. $\mathbb{R} \subseteq \mathbb{C}$ are both fields. But $\mathbb{C}$ is a 2-dimensional vector space over $\mathbb{R}$ as we can take the basis $\{1, i\}$. So $\mathbb{C}/\mathbb{R}$ is a finite extension of degree 2. But $\mathbb{Q}/\mathbb{R}$ is an infinite field extension. On the other hand let $\alpha \in \mathbb{R}$. Then consider the field $\mathbb{Q}(\alpha)$. Now degree of $[\mathbb{Q}(\alpha)\colon \mathbb{Q}]$ depends on what kind of element α is. If $\alpha = \sqrt{2}$ then $[\mathbb{Q}(\sqrt{2})\colon \mathbb{Q}] = 2$ but if $\alpha = \pi$ then $[\mathbb{Q}(\pi)\colon \mathbb{Q}] = \infty$.

Definition 1.1.1: Generated Subfields

Let K/F be a field extension and $S \subseteq K$.

(i) The ring generated by F and S, *written $F[S]$, is the intersection of all subrings of K containing F and S.*

(ii) The field generated by F and S, *written $F(S)$, is the intersection of all subfields of K containing F and S.*

If $S = \{a_1, \ldots, a_n\}$ is finite we write $F[a_1, \ldots, a_n]$ and $F(a_1, \ldots, a_n)$. We call $F(\alpha_1, \ldots, \alpha_n)$ finitely generated *field. For any field extension K/F if there exists an element $\alpha \in K$ such that $K = F(\alpha)$ then such extension is called* simple extension.

From this definition it easily follows that for any $S \subseteq K$, $F(S)$ is the smallest subfield of K containing F and α.

Let K/F be a field extension. For any $\alpha \in K$ consider the map $\mathrm{ev}_\alpha : F[X] \to K$ defined by $f(X) \mapsto f(\alpha)$. Hence, ev_α is a ring homomorphism.

Observation 1.1. *For any $\alpha \in K$, $\mathrm{Im}(\mathrm{ev}_\alpha)$ is a subring of K and ev_α is F-vector space homomorphism.*

Now we can use this map ev_α to give descriptions of the elements of $F[\alpha]$ and $F(\alpha)$.

Lemma 1.1.1

Let K be a field extension of F and let $a \in K$. Then

$$F[a] = \{f(a) : f(x) \in F[x]\} \quad \textit{and} \quad F(a) = \left\{\frac{f(a)}{g(a)} : f, g \in F[x],\ g(a) \neq 0\right\}.$$

Moreover, $F(a)$ is the quotient field of $F[a]$.

Proof: The evaluation map ev_a is a ring homomorphism with image $\{f(a) : f \in F[x]\}$/ So this set is a subring of K. Any subring of K containing F and a must contain $f(a)$, for all $f \in F[X]$. Therefore, $F[a] = \{f(a) : f \in F[x]\}$. The fraction field of $F[a]$ is $\{f(a)/g(a)\colon g(a) \neq 0\}$; it is contained in any subfield of K containing $F[a]$, hence equals $F(a)$. ■

We call this procedure *adjoining* α to F. Using similar arguments we can describe the ring $F[a_1, \ldots, a_n]$ and the field $F(a_1, \ldots, a_n)$ where K/F is a field extension and $a_1, \ldots, a_n \in K$. So we have the following theorem.

Theorem 1.1.2

Let K be a field extension of F and let $a_1, \ldots, a_n \in K$. Then

$$F[a_1, \ldots, a_n] = \{f(a_1, \ldots, a_n) : f \in F[x_1, \ldots, x_n]\},$$
$$F(a_1, \ldots, a_n) = \left\{\frac{f(a_1, \ldots, a_n)}{g(a_1, \ldots, a_n)} : f, g \in F[x_1, \ldots, x_n],\ g(a_1, \ldots, a_n) \neq 0\right\}.$$

Moreover, $F(a_1, \ldots, a_n)$ is the quotient field of $F[a_1, \ldots, a_n]$.

Now we can also define fields generated by adjoining arbitrary number of elements. In this case we can give the description of the new field in terms of all finite extension of fields created by adjoining finitely many elements.

Theorem 1.1.3

Let K be a field extension of F and let $S \subseteq K$ be any subset. If $\alpha \in F(S)$, then $\alpha \in F(a_1, \ldots, a_n)$ for some $a_1, \ldots, a_n \in S$. Therefore,

$$F(S) = \bigcup \{F(a_1, \ldots, a_n) \colon a_1, \ldots, a_n \in S\},$$

where the union is over all finite subsets of S.

Proof: Let $U := \cup \{F(a_1, \ldots, a_n) \colon a_1, \ldots, a_n \in S\}$. Each $F(a_1, \ldots, a_n)$ contains F and lies in $F(S)$ where $a_1, \ldots, a_n \in S$. So the union U satisfies $U \subseteq F(S)$. The set U contains F and S. If $\alpha \in F(a_1, \ldots, a_n)$ and $\beta \in F(b_1, \ldots, b_m)$, then $\alpha \pm \beta$, $\alpha\beta$, α/β (if $\beta \neq 0$) are all in $F(a_1, \ldots, a_n, b_1, \ldots, b_m) \subseteq U$. Therefore, U is a field. Being the smallest subfield containing F and S, it equals $F(S)$. ■

1.1.2 Algebraic Elements and Minimal Polynomials

Definition 1.1.2: Algebraic and Transcendental Elements

Let K be a field extension of F. An element $\alpha \in K$ is algebraic over F *if there is a nonzero polynomial $f(X) \in F[X]$ with $f(\alpha) = 0$. If α is not algebraic over F, then α is called* transcendental over F. *If every element of K is algebraic over F, then K is said to be* algebraic over F, *and K/F is called an* algebraic extension.

Hermite (1873) proved that e is transcendental over $\mathbb{Q}$ and Lindemann (1882) proved the same for π. On the other hand, $\sqrt[n]{r} \in \mathbb{R}$ is algebraic over $\mathbb{Q}$ for any $r \in \mathbb{Q}$ (root of $x^n - r$), as is $i = \sqrt{-1}$ (root of $x^2 + 1$).

Definition 1.1.3: Minimal Polynomial

If α is algebraic over a field F, the minimal polynomial *of α over F is the monic polynomial $p(X)$ of least degree in $F[X]$ for which $p(\alpha) = 0$; it is denoted* $\min(F, \alpha)$. *Equivalently,* $\min(F, \alpha)$ *is the monic generator of the kernel of the evaluation homomorphism* ev_α.

The minimal polynomial depends on the base field. For instance, $\min(\mathbb{Q}, i) = x^2 + 1$ while $\min(\mathbb{C}, i) = x - i$, since $i \in \mathbb{C}$ is already in the base field.

The minimal polynomial of an element and the degree of a field extension are two of the most basic tools we shall use. The following proposition gives a relation between these objects.

Theorem 1.1.4

Let K/F be a field extension and let $\alpha \in K$ be algebraic over F. If $p(X) = \min(F, \alpha)$, then:

(i) *$p(X)$ is irreducible over F.*

(ii) *If $g(X) \in F[X]$, then $g(\alpha) = 0$ if and only if $p(X) \mid g(X)$.*

(iii) *If $n = \deg(\min(F, \alpha))$, then the elements $1, \alpha, \ldots, \alpha^{n-1}$ form a basis for $F(\alpha)$ over F. So $[F(\alpha) \colon F] = \deg(\min(F, \alpha)) < \infty$. Moreover, $F(\alpha) = F[\alpha]$.*

(iv) *$F(\alpha) \cong F[X] \,/\, \langle p(X) \rangle$.*

Proof: The *first isomorphism theorem* gives $F[\alpha] \cong F[X] \,/\, \langle p(X) \rangle$. Since $F[\alpha] \subseteq K$ is an integral domain, $\langle p(X) \rangle$ is a prime ideal. Now in an PID every nonzero prime is maximal, so $F[X] \,/\, \langle p(X) \rangle$ is a field and $F[\alpha] = F(\alpha)$. This gives (i) and (iv). Part (ii) follows from $\ker(\mathrm{ev}_\alpha) = \langle p(X) \rangle$.

For (iii): any $b \in F(\alpha) = F[\alpha]$ satisfies $b = g(\alpha)$. Dividing gives $g = qp + r$ with $\deg r < n$ and $q, r \in F[X]$. So $b = r(\alpha)$ lies in the span of $\{1, \alpha, \ldots, \alpha^{n-1}\}$. Therefore, the set $\{1, \alpha, \ldots, \alpha^{n-1}\}$ spans $F(\alpha)$ as a vector space. If $a_0 + a_1 \cdot \alpha + \cdots + a_{n-1} \cdot \alpha^{n-1} = 0$ where $a_0, \ldots, a_{n-1} \in F$ then the polynomial $f(X) = \sum_{i=0}^{n-1} a_i X^i$ is in $\ker(\mathrm{ev}_\alpha)$. Hence, either $f \equiv 0$ or $p \mid f$ by part (ii). But $\deg(f) < n$. So $f \equiv 0$. Thus, $1, \alpha, \ldots, \alpha^{n-1}$ forms a basis of $F(\alpha)$ over F. ■

The part (iii) of above theorem gives us a basis of $F(\alpha)$ over F using only α. So we get the following observation:

Observation 1.2. *If K/F is a field extension and $\alpha \in K$ is algebraic over F where $p = \min(F, \alpha)$ then $[F(\alpha) \colon F] = \deg(p)$.*

Notice that if $\alpha, \beta \in K$ are both algebraic over F and $\min(F, \alpha) = \min(F, \beta) = p$ i.e., their minimal polynomials are same then by the above theorem

$$F(\alpha) \cong F[X] \Big/ \langle p \rangle \cong F(\beta)$$

Therefore we have the observation:

Observation 1.3. *If K/F is a field extension and $\alpha, \beta \in K$ are both algebraic over F, and they have the same minimal polynomials then $F(\alpha) \cong F(\beta)$ with the isomorphism $\varphi : \alpha \mapsto \beta$.*

1.1.3 Finite and Algebraic Extensions

Lemma 1.1.5

If $[K : F] < \infty$, then K is algebraic and finitely generated over F.

Proof: Let $\{\alpha_1, \ldots, \alpha_n\}$ be an F-basis of K. Then for any element $\alpha \in K$, there exists $a_1, \ldots, a_n$ such that $\alpha = a_1 \cdot \alpha_1 + \cdots + a_n \cdot \alpha_n$. So certainly we have $K = F(\alpha_1, \ldots, \alpha_n)$ and hence K is finitely generated over F. Since $[K \colon F] = n$, for any $\alpha \in K$, the $n + 1$ elements $1, \alpha, \ldots, \alpha^n$ span only an n-dimensional space. So they are F-linearly dependent. Thus, there exists $\beta_0, \ldots, \beta_n \in F$ such that $\sum_{i=0}^{n} \beta_i \cdot \alpha^i = 0$. So we get a polynomial $f(X) = \sum_{i=0}^{n} \beta_i \cdot X^i$ and $f(\alpha) = 0$. Therefore, α is algebraic over F. ■

Let K/F is a finite extension and $K = F(\alpha_1, \ldots, \alpha_n)$. Then we can break up the extension K/F into collection of subextensions. Let $L_i = F(\alpha_1, \ldots, \alpha_i)$. Set $L_0 = F$. Then basically $L_{i+1} = L_i(\alpha_{i+1})$. And we have

$$F = L_0 \subseteq L_1 \subseteq \cdots \subseteq L_{n-1} \subseteq L_n = K$$

We will show a theorem which relates the degree of the extension K/F and each subextensions.

Theorem 1.1.6

Let $F \subseteq L \subseteq K$ be fields. Then $[K : F] = [K : L] \cdot [L : F]$.

Proof: Let $\{a_i\}_{i\in I}$ be an F-basis of L and $\{b_j\}_{j\in J}$ an L-basis of K. We show $\{a_i \cdot b_j\}_{I\times J}$ is an F-basis of K. For any $\gamma \in K$, write $\gamma = \sum_j \alpha_j b_j$ with $\alpha_j \in L$. Now each $\alpha_j = \sum_i \beta_{ij} a_i$ with $\beta_{ij} \in F$; then $\gamma = \sum_{i,j} \beta_{ij} a_i b_j$. Therefore, the set $\{a_i \cdot b_j\}_{I\times J}$ spans K over F.

If $\sum_{i,j} \beta_{ij} a_i b_j = 0$ for some non-trivial linear combinations where $\beta_{i,j} \in F$ for all $i \in I$, $j \in J$. Now

$$\sum_{i,j} \beta_{ij} a_i b_j = \sum_j \Big(\sum_i \beta_{ij} a_i\Big) b_j = 0.$$

Since $\{b_j\}_{j\in J}$ is a basis of K over L, each $\sum_i \beta_{ij} a_i = 0$. And by F-independence of $\{a_i\}$, each $\beta_{ij} = 0$. So $\{a_i \cdot b_j\}_{I\times J}$ form an F-basis of K. Therefore,

$$[K : F] = |\{a_i \cdot b_j\}_{I\times J}| = |\{a_i : i \in I\}| \cdot |\{b_j : j \in J\}| = [K : L] \cdot [L : F]$$

So we have the theorem. ■

Lemma 1.1.7

Let $K = F(X)$ be the field of rational functions in X over F. If $u \in K$ with $u \notin F$ let $u = f/g$ where $f, g \in F[X]$ and $\gcd(f, g) = 1$. Let $L = F(u)$. Then

$$[K : L] = \max\{\deg(f), \deg(g)\}$$

Proof: By Theorem 1.1.4 we need to find the minimal polynomial of X over L to determine $[K : L]$. Consider the polynomial $p(Y) = u \cdot g(Y) - f(Y) \in L[Y]$. Then X is a root of p. Therefore, X is algebraic over L and so $[K : L] < \infty$ and $K = L(X)$.

Notice that $\deg(p) = \max\{\deg(f), \deg(g)\}$. Let $f = \sum_{i=0}^{n} a_i X^i$ and $g = \sum_{j=0}^{m} b_i \cdot X^i$. We will show that p is irreducible over L. Now the element u cannot be algebraic over k otherwise by Theorem 1.1.6 we have $[K : F] = [K : L] \cdot [L : F] < \infty$, and we know $F(X)$ is an infinite extension of F. So u is transcendental over F and hence $F[u] \cong F[X]$. Now observe that the polynomial $p \in F[Y][u] \subseteq F(Y)[u]$ which is of degree 1. Hence, p is irreducible over $k(Y)$. Now $p \in k[Y][u] = k[u][Y]$ and therefore p is irreducible over $k(u)$. Hence, p is the minimal polynomial of X over F. So we have the lemma. ■

Note:-

In the proof of Theorem 1.1.6 the proof do not assume that the field extensions needs to be finite.

We will now prove a converse theorem of Lemma 1.1.5.

Theorem 1.1.8

Let K/F be a field extension. If $\alpha_1, \ldots, \alpha_n \in K$ are algebraic over F, then

$$[F(\alpha_1, \ldots, \alpha_n) : F] \leq \prod_{i=1}^{n} [F(\alpha_i) : F].$$

Proof: We will prove this by induction on n. The base case, $n = 1$ follows from the Theorem 1.1.6. So suppose this is true for $n - 1$. Take $L = F(\alpha_1, \ldots, \alpha_{n-1})$. Then by induction hypothesis $[L : F] \leq \prod_{i=1}^{n-1} [F(\alpha_i) : F]$. Since α_n is

algebraic over F and hence over L, with $\min(L, \alpha_n) \mid \min(F, \alpha_n)$ in $L[X]$, we have $[L(\alpha_n) : L] \leq [F(\alpha_n) : F]$. Therefore, $[F(\alpha_1, \ldots, \alpha_n) : L] \leq [F(\alpha_n) : F]$. So by Theorem 1.1.6 we have

$$[F(\alpha_1, \ldots, \alpha_n) : F] = [L(\alpha_n) : L] \cdot [L : F] \leq [F(\alpha_n) : F] \cdot \prod_{i=1}^{n-1} [F(\alpha_i) : F].$$

So we have the theorem. ■

The inequality above can be strict in some cases. For example let $a = \sqrt[4]{2}$ and $b = \sqrt[4]{18}$. Then both $[\mathbb{Q}(a) : \mathbb{Q}] = [\mathbb{Q}(b) : \mathbb{Q}] = 4$ as $X^4 - 2$ and $X^4 - 18$ are both irreducible polynomials over $\mathbb{Q}$. But $[\mathbb{Q}(\sqrt[4]{2}, \sqrt{3}) : \mathbb{Q}] = 8$ and $\sqrt[4]{18} \in \mathbb{Q}(\sqrt[4]{2}, \sqrt{3})$. Hence, in this the inequality doesn't follow.

So from, and Theorem 1.1.8 we have the following criterion for an element being algebraic over a field.

Corollary 1.1.9

Let K/F be a field extension and $\alpha \in K$. Then α is algebraic over F if and only if $[F(\alpha) : F] < \infty$. Moreover, K is algebraic over F if $[K : F] < \infty$.

Now we can extend Theorem 1.1.8 to the case of fields generated by arbitrary number of elements or extending field.

Theorem 1.1.10

Let K be a field extension of F and $X \subseteq K$ such that each element of S is algebraic over F. Then $F(S)/F$ is algebraic. If $|S| < \infty$, then $[F(S) : F] < \infty$.

Proof: For any $\alpha \in F(S)$. Theorem 1.1.3 gives $\alpha \in F(a_1, \ldots, a_m)$ for some $a_1, \ldots, a_m \in S$. By Theorem 1.1.8, $F(a_1, \ldots, a_m)$ is algebraic over F. So α is algebraic over F by Theorem 1.1.5. Since α is any arbitrary element of $F(S)$, $F(S)$ is algebraic over F. If S is finite, then $[F(S) : F] < \infty$ by Theorem 1.1.8. ■

We will now show that the property of being algebraic is a transitive property. In the finite extensions case this property easily follows from Theorem 1.1.6 and Corollary 1.1.9. We will show this for general extensions below.

Theorem 1.1.11 Transitivity of Algebraic Extensions

Let $F \subseteq L \subseteq K$ be fields. If L/F and K/L are algebraic, then K/F is algebraic.

Proof: Let $\alpha \in K$. Since K/L is algebraic, let $f(X) = \sum_{i=0}^{d} a_i X^i$ where each $a_i \in L$ is the minimal polynomial of α over L. Since L/F is algebraic and $f \in L[X]$, each a_i is algebraic over F. Consider the field $L' = F(a_0, \ldots, a_d)$. So by Theorem 1.1.8 the extension L'/F is finite i.e., $[L' : F] < \infty$. Since $f \in L'[X]$ and $p(\alpha) = 0$, the element α is algebraic over L' and $[L'(\alpha) : L'] < \infty$. By the tower law,

$$[L'(\alpha) : F] = [L'(\alpha) : L'] \cdot [L' : F] < \infty,$$

Since $F(\alpha) \subseteq L'(\alpha)$, α is algebraic over F. Since α is any arbitrary element of K, K/F is algebraic. ■

Definition 1.1.4: Algebraic Closure in K

Let K be a field extension of F. The set

$$\{\, a \in K : a \text{ is algebraic over } F \,\}$$

is called the algebraic closure of F in K.

Corollary 1.1.12

Let K be a field extension of F and let L be the algebraic closure of F in K. Then L is a field, and therefore is the largest algebraic extension of F contained in K.

Proof: Let $a, b \in L$. Then both are algebraic over F by Theorem 1.1.8 and hence $F(a, b)$ is algebraic over F and $F(a, b) \subseteq L$. Since $a \pm b, a \cdot b, a/b$ (if $b \neq 0$) all are in $F(a, b) \subseteq L$, L is closed under the field operation. So L is a subfield of K. Since each element of K which is algebraic over F is contained in L, L is the largest algebraic extension of F contained in K. ■

1.1.4 Splitting Field

Definition 1.1.5: Splits over K

If K is an extension field of F and if $f(X) \in F[X]$, then f splits over K *if $f(X) = a \prod_i (X - \alpha_i) \in K[X]$ for some $\alpha_1, \ldots, \alpha_n \in K$ and $a \in F$. In other words, f splits over K if f factors completely into linear factors in $K[X]$.*

In order to talk about roots of a given polynomial, we need to have extension fields that contain the roots of the polynomial. The next theorem shows that for any $f \in F[X]$, there is a finite extension of F over which f splits.

Theorem 1.1.13

Let $f(x) \in F[x]$ have degree n. There is an extension field K of F with $[K : F] \leq n$ such that K contains a root of f. In addition, there is a field L containing F with $[L : F] \leq n!$ such that f splits over L.

Proof: Let $p(X)$ is an irreducible factor of $f(X)$ in $F[X]$ and let $K = F[X] / \langle p(X) \rangle$. Thus, K contains a root of $p(X)$ say α. Therefore, α is also a root of $f(X)$. Since $[K\colon F] = \deg(p) \leq n$ we have the first part.

For the second part we will use induction on n. From the first part now we have a field $K \supseteq F$ such that K contains a root $\alpha of F$. So f factors into $f(X) = (X - \alpha)g(X)$ in $K[X]$ where $g(X) \in K[X]$. Now $\deg(g) = n - 1$. So by induction hypothesis there exists a finite extension L of K such that $[L\colon K] \leq (n-1)!$ and g splits over L. That means f splits over L and $[L\colon F] = [L\colon K] \cdot [K\colon F] \leq n!$. So we have the theorem. ■

Definition 1.1.6: Splitting Field

Let K be an extension of F and let $f \in F[X]$.

(i) K is called the splitting field of f over F *if and only if f splits over K and $K = F(\alpha_1, \ldots, \alpha_n)$ where $\alpha_1, \ldots, \alpha_n$ are exactly the roots of f.*

(ii) If S is a set of non-constant polynomials over F then K is the splitting field of S over F *if and and only if each $f \in S$ splits over K and $K = F(S_R)$ where S_R is the set of all roots of all $f \in S$.*

So Theorem 1.1.13 yields immediately the existence of splitting fields for a finite set of polynomials. We have proved the existence of splitting fields for finite sets of polynomials. What about infinite sets? Suppose that K is a splitting field over F of the set of all non-constant polynomials over F. We do not know yet that such a field exists or not.

Lemma 1.1.14

If K is a field, then the following statements are equivalent:

(i) There are no algebraic extensions of K other than K itself.

(ii) There are no finite extensions of K other than K itself.

(iii) If L is a field extension of K, then $K = \{a \in L : a \text{ is algebraic over } K\}$.

(iv) Every $f(x) \in K[x]$ splits over K.

(v) Every $f(x) \in K[x]$ has a root in K.

(vi) Every irreducible polynomial over K has degree 1.

Proof: We will show the implications in a circular order, (i) $\implies$ (ii) $\implies$ (iii) $\implies$ (iv) $\implies$ (v) $\implies$ (vi) $\implies$ (i).

(i) $\implies$ (ii): This is clear since any finite extension is algebraic.

(ii) $\implies$ (iii): Let $a \in L$ is algebraic over K. Then $K(a)$ is a finite extension of K. But there are no finite extensions of K other than itself. So $K(a) = K$. Thus, $a \in K$.

(iii) $\implies$ (iv): Let $f \in K[X]$ and L be the splitting field of f over K. Since L is algebraic over K we have $L = K$ i.e., f splits over K.

(iv) $\implies$ (v): This is obviously true.

(v) $\implies$ (vi): Let $f \in K[X]$ is irreducible. Therefore, f has a root in K, so f has a linear factor. But f is irreducible. Hence, $\deg(f) = 1$.

(vi) $\implies$ (i): Let L be an algebraic extension of K. Let $a \in L$ and $p = \min(K, a)$. Then $\deg(p) = 1$. So $[K(a) : K] = 1$ and hence $a \in K$. So $L = K$. ■

Definition 1.1.7: Algebraically Closed Field

If any field K satisfies the equivalent conditions of Lemma 1.1.14 then K is said to be algebraically closed. *If K is an algebraic extension of F then K is said to be* algebraic closure *of F.*

We haven't showed the existence of such theorem or even every field as an algebraic closure. Consider the field $\mathbb{C}$. This field is algebraically closed. But we will not prove if every field has an algebraic closure. We will just state the theorem and move on.

Theorem 1.1.15

Let F is a field. Then F has an algebraic closure.

The existence of an algebraic closure yields immediately the existence of a splitting field for an arbitrary set of non-constant polynomials.

Corollary 1.1.16

Let F be a field and S be a set of non-constant polynomials over F. Then S has a splitting field over F.

Proof: Let K be an algebraic closure of F. Then each $f(x) \in S$ splits over K. Let X be the set of roots of all $f \in S$. Then $F(X) \subseteq K$ is a splitting field for S over F, since each f splits over $F(X)$ and this field is generated by the roots of all the polynomials from S. ■

So using the corollary we can immediately relate algebraic closure with splitting fields.

Corollary 1.1.17

If F is a field then the splitting field of the set of all non-constant polynomials over F is an algebraic closure of F.

But now comes another question. Can one field have multiple algebraic closures? Or even can we have more than one splitting fields? The answer is splitting fields are unique, and hence algebraic closures are also unique. Below we share the theorem which proves the above. We will not prove it because its beyond our scope of study. But for further reading and understanding the reader can look [Mor96].

Theorem 1.1.18 Isomorphism Extension Theorem, [Mor96, Chapter 1]

Let $\sigma : F \longrightarrow F'$ be a field isomorphism. Let $S = \{f_i(x)\}$ be a set of polynomials over F, and let $S' = \{\sigma(f_i)\}$ be the corresponding set over F'. Let K be a splitting field for S over F, and let K' be a splitting field for S' over F'. Then there is an isomorphism $\tau : K \longrightarrow K'$ with $\tau|_F = \sigma$. Furthermore, if $\alpha \in K$ and α' is any root of $\sigma(\min(F, \alpha))$ in K', then τ can be chosen so that $\tau(\alpha) = \alpha'$.

The isomorphism extension theorem immediately settles the question of uniqueness of splitting fields and algebraic closures: any two splitting fields of the same set of polynomials over F must be isomorphic, since the identity on F extends to an isomorphism between them. As a special case, algebraic closures are unique up to F-isomorphism.

Corollary 1.1.19

Let F be a field, and let S be a subset of $F[x]$. Any two splitting fields of S over F are F-isomorphic. In particular, any two algebraic closures of F are F-isomorphic.

Proof: For the proof of the first statement, the isomorphism extension theorem gives an isomorphism extending id on F between any two splitting fields of S. The second statement follows from the first, since any algebraic closure of F is a splitting field of the set of all non-constant polynomials in $F[x]$. ■

As a corollary to the existence and uniqueness of algebraic closures, we can prove that any algebraic extension of a field F can be viewed as living inside a fixed algebraic closure of F.

Corollary 1.1.20

Let F be a field, and let N be an algebraic closure of F. If K is an algebraic extension of F, then K is isomorphic to a subfield of N.

Proof: Let M be an algebraic closure of K. By Theorem 1.1.11, M is algebraic over F; hence, M is also an algebraic closure of F. Therefore, by the previous corollary, $M \cong N$. If $f : M \to N$ is an F-isomorphism, then $f(K)$ is a subfield of N isomorphic to K. ■

§ 1.2 Some Properties of Finite Fields

In this section we describe fields of finite order. The characteristic of a finite field cannot be zero so it is a prime number. Therefore, if K is a finite field and $\mathrm{char}(K) = p$ then $\mathbb{F}_p \subseteq K$. Since K is finite it will be a finite dimensional vector space over $\mathbb{F}_p$. So K is isomorphic to $\mathbb{F}_p^r$ where $r = [K\colon \mathbb{F}_p]$ as a $\mathbb{F}_p$-vector space which contains p^r elements. So order of a finite field is always a prime power. We write $\mathbb{F}_q := K$ where $q = p^r$. Below we list the main facts of finite fields without proof. If the reader is interested he can look into the book [Art11].

Theorem 1.2.1 Properties of Finite Field

Let p be a prime integer, and let $q = p^r$ be a positive power of p.

(i) Let K be a field of order q. The elements of K are roots of the polynomial $X^q - X$.

(ii) The irreducible factors of the polynomial $X^q - X$ over the field $\mathbb{F}_p$ are the irreducible polynomials in $\mathbb{F}_p[X]$ whose degrees divide r.

(iii) Let K be a field of order q. The multiplicative group K^* of nonzero elements of K is a cyclic group of order $q - 1$.

(iv) There exists a field of order q, and all fields of order q are isomorphic.

(v) A field of order p^r contains a subfield of order p^k if and only if k divides r.

Observation 1.4. *Since for every $r \in \mathbb{N}$ there exists an element $\alpha \in \mathbb{F}_{q^r}$ such that $\mathbb{F}_{q^r}^*$ is generated by α then $\mathbb{F}_{q^r} = \mathbb{F}_q(\alpha)$. Hence, every finite extensions of finite fields are simple extensions.*

Since finite fields are finite extensions of the base prime field $\mathbb{F}_p$ we can obtain irreducible polynomials for every positive integer degree.

Corollary 1.2.2

For every $d \in \mathbb{N}$ there exists an irreducible polynomial of degree d over the finite field $\mathbb{F}_p$.

Proof: By the part (iv) of the above theorem here is a field K of order p^d. Its degree $[K\colon \mathbb{F}_p] = d$. By part (iii) there is an element $\alpha \in K$ such that α generates the cyclic group K^*. So $K = \mathbb{F}_p(\alpha)$. Therefore, α is a root of a d-degree irreducible polynomial over $\mathbb{F}_p$. ■

Definition 1.2.1: Primitive Element

Let K be finite field over $\mathbb{F}_p$. Then K has an element α which generates K/F i.e., $K = \mathbb{F}_p(\alpha)$. Then α is called the primitive element *of K.*

Let f is an irreducible polynomial over the finite field $\mathbb{F}_q$ and let α be a root of f in the splitting field of f over $\mathbb{F}_q$. Therefore, $f = \min(\mathbb{F}_q, \alpha)$. Now suppose $h \in \mathbb{F}_q[X]$ such that $h(\alpha) = 0$. So by Theorem 1.1.4, $f \mid h$. Like Theorem 1.2.1(ii) we will show that we can similar result for any general finite field.

Lemma 1.2.3

Let $f \in \mathbb{F}_q[X]$ be an irreducible polynomial of degree d. Then $f(X)$ divides $X^{q^n} - X$ if and only if $d \mid n$.

Proof: Suppose $f \mid X^{q^n} - X$. Let α be a root of f in the splitting field. Then $\alpha^{q^n} = \alpha$ and so $\alpha \in \mathbb{F}_{q^n}$. Now $\mathbb{F}_q \subseteq \mathbb{F}_q(\alpha) \subseteq \mathbb{F}_{q^n}$. Since $[\mathbb{F}_q(\alpha) : \mathbb{F}_q] = d$ and $[\mathbb{F}_{q^n} : \mathbb{F}_q] = n$. So by Theorem 1.1.6, $d \mid n$.

Let $d \mid n$. Then $\mathbb{F}_{q^d} \subseteq \mathbb{F}_{q^n}$. If α is a root of f in the splitting field of f over $\mathbb{F}_q$ then $\mathbb{F}_q(\alpha) = \mathbb{F}_{q^d} \subseteq \mathbb{F}_{q^n}$. Therefore, α is a root of $X^{q^n} - X$. Since f is the minimal polynomial of α and $X^{q^n} - X \in \mathbb{F}_q$, $f \mid X^{q^n} - X$. ■

Theorem 1.2.4

If f is an irreducible polynomial of $\mathbb{F}_q[X]$ of degree d then f has a root $\alpha \in \mathbb{F}_{q^d}$. Furthermore, all roots of f are given by d distinct elements $\alpha, \alpha^q, \alpha^{q^2}, \ldots, \alpha^{q^{d-1}}$ of $\mathbb{F}_{q^d}$.

Proof: Let $f(X) = \sum_{i=0}^{d} a_i X^i$. Then

$$f\left(\alpha^{q^j}\right) = \sum_{i=0}^{d} a_i \cdot \left(\alpha^{q^j}\right)^i = \sum_{i=0}^{d} a_i^{q^j} \cdot \left(\alpha^i\right)^{q^j} = \left(\sum_{i=0}^{d} a_i \cdot \alpha^i\right)^{q^j} = f^{q^j}(\alpha) = 0$$

Since $\alpha^{q^d} = \alpha$ the roots of f are $\alpha, \alpha^q, \ldots, \alpha^{q^{d-1}}$. ■

This introduces us to look at roots and field automorphisms in a new way. Let F, F' are two fields and $\varphi : F \to F'$ is a homomorphism. Then there is an induced homomorphism $\varphi : F[X] \to F'[X]$ given by

$$\varphi\left(\sum_{i=0}^{n} a_i \cdot X^i\right) = \sum_{i=0}^{n} \varphi(a_i) \cdot X^i$$

So if $f = \sum_{i=0}^{n} a_i X^i$ factors into $a(X - \alpha_1) \cdots (X - \alpha_n)$ then $\varphi(f) = \varphi(a)(X - \varphi($
$alpha_1)) \cdots (X - \varphi(\alpha_n))$. The *Frobenius map*, $\sigma : \alpha \mapsto \alpha^q$ for any element of the extended field $\mathbb{F}_{q^d}$ over $\mathbb{F}_q$ is a $\mathbb{F}_q$-automorphism meaning that automorphism fixes the elements of $\mathbb{F}_q$. The above theorem tells us if α is a root of f which is an irreducible polynomial over $\mathbb{F}_q$ then $\sigma(\alpha)$ is also a root.

Now by the above theorem if f is irreducible over $\mathbb{F}_q$ of degree d then $\mathbb{F}_{q^d}$ contains a root α of f. Then $\alpha, \alpha^q, \ldots, \alpha^{q^{d-1}}$ all are in $\mathbb{F}_{q^d}$. Therefore, f splits in $\mathbb{F}_{q^d}$. Since f is the minimal polynomial of α, $\mathbb{F}_{q^d} = \mathbb{F}_q[X] / \langle f \rangle$. So $\mathbb{F}_{q^d}$ is the splitting field of f. Since all finite fields of size q^d are isomorphic for any other degree d irreducible polynomial $g \in \mathbb{F}_q[X]$, if β is a root of g then

$$\mathbb{F}_q(\alpha) \cong \mathbb{F}_q[X] \big/ \langle f \rangle \cong \mathbb{F}_{q^d} \cong \mathbb{F}_q[X] \big/ \langle g \rangle \cong \mathbb{F}_q(\beta)$$

So the splitting fields of f and g are isomorphic. So we have the following corollary.

Corollary 1.2.5

Let f is an irreducible polynomial of $\mathbb{F}_q[X]$ of degree d. Then the splitting field of f over $\mathbb{F}_q$ is given by $\mathbb{F}_{q^d}$. Moreover, any two irreducible polynomials $f, g \in \mathbb{F}_q[X]$ of same degree has isomorphic splitting fields.

Now we have heard the term conjugate of an element in complex numbers. The notion of conjugates is more general than just in complex numbers.

Definition 1.2.2

Let $\mathbb{F}_{q^d}$ be an extension of $\mathbb{F}_q$ and $\alpha \in \mathbb{F}_{q^d}$. Then the elements $\alpha, \alpha^q, \ldots, \alpha^{q^{d-1}}$ are called the conjugates *of α.*

The conjugates of $\alpha \in \mathbb{F}_{q^m}$ with respect to $\mathbb{F}_q$ are distinct if and only if the minimal polynomial of α over $\mathbb{F}_q$ has degree m. Otherwise, the degree d of this minimal polynomial is a proper divisor of m, and then the conjugates of α with respect to $\mathbb{F}_q$ are the distinct elements $\alpha, \alpha^q, \ldots, \alpha^{q^{d-1}}$, each repeated m/d times.

Lemma 1.2.6

For any $\alpha \in \mathbb{F}_q^*$ the conjugates with respect to any subfield of $\mathbb{F}_q$ have the same order in the group $\mathbb{F}_q^*$.

Proof: Let $\mathbb{F}_{q'}$ be a subfield of $\mathbb{F}_q$ where $q'^r = q$. Since $\mathbb{F}_q^*$ is a cyclic group of order $q-1$ and $\gcd(q'^i, q-1) = 1$ for all $0 \le i \le r-1$. Therefore, the order of the conjugates of α has the same order as α in the group $\mathbb{F}_q^*$. ■

§ 1.3 Trace and Norm

Let V is an n-dimensional vector space over K where K is a field. Let $\varphi : V \to V$ is any linear operator. For any basis of V we can generate the matrix of φ with respect to that basis. This is called the *matrix representation* of φ and denoted by $[\varphi]$.

We now turn to field extensions. For a finite extension of field L/K we associate each element α of L the K-linear transformation $m_\alpha : L \to L$ where $m_\alpha(x) = \alpha \cdot x$ for any $x \in L$. Now m_α is indeed a linear transformation from L to L as

$$m_\alpha(x+y) = \alpha(x+y) = \alpha \cdot x + \alpha \cdot y = m_\alpha(x) + m_\alpha(y) \qquad m_\alpha(c \cdot x) = \alpha(c \cdot x) = c(\alpha \cdot x) = c \cdot m_\alpha(x)$$

for all $x, y \in L$ and $c \in K$. Now by choosing a K-basis of L we can create a matrix representation for m_α which is denoted by $[m_\alpha]$. So for any $c \in K$ the matrix $[m_c]$ is cI_k where $[L\colon K] = n$.

Definition 1.3.1: Trace and Norm of α

For any $\alpha \in L$ where L/K be a finite extension the trace and norm of α from L to K are the trace and determinant of a matrix representation of the K-linear map m_α and denoted as

$$\mathrm{Tr}_{L/K}(\alpha) = \mathrm{Tr}([m_\alpha]) \in K, \qquad \mathrm{N}_{L/K}(\alpha) = \det([m_\alpha])$$

For any finite field $\mathbb{F}_q$ if $\mathbb{F}_p$ is the base prime field of $\mathbb{F}_q$ then for any $\alpha \in \mathbb{F}_q$ we denote the trace and norm of α from $\mathbb{F}_q$ to $\mathbb{F}_p$ by $\mathrm{Tr}_{\mathbb{F}_q}(\alpha)$ and $\mathrm{N}_{\mathbb{F}_q}(\alpha)$, and we call it the *absolute trace* and *absolute norm* of α respectively.

Observation 1.5. *For any $\alpha \in K$, where $[L\colon K] = n$, $\mathrm{Tr}_{L/K}(\alpha) = n \cdot \alpha$ and $\mathrm{N}_{L/K}(\alpha) = \alpha^n$. In particular $\mathrm{Tr}_{L/K}(1) = [L\colon K]$ and $\mathrm{N}_{L/K}(1) = 1$.*

Note:-

There is an alternate definition of trace and norm in terms of sum and product of field embeddings of L into an algebraic closure of K. But those definitions is hard to work with and also beyond our scope. So interested readers can look into [Lan02, Chapter 1]

Lemma 1.3.1

Let $\alpha, \beta \in L$.

(i) If $\alpha \neq \beta$ then $m_\alpha \neq m_\beta$.

(ii) As functions from $L \to L$

$$m_{\alpha+\beta} = m_\alpha + m_\beta \quad \text{and} \quad m_{\alpha\beta} = m_\alpha \circ m_\beta$$

and m_1 is the identity map $L \to L$.

Proof: Since $m_\alpha(1) = \alpha$ we can recover the element α uniquely from the mapping m_α. Therefore, the map, $\varphi : L \to \{m_\alpha : \alpha \in L\}$ where $\varphi(\alpha) = m_\alpha$ is a bijection. And hence if $\alpha, \beta \in L$ and $\alpha \neq \beta$ then $m_\alpha \neq m_\beta$.

Now for any $x \in L$ we have

$$m_{\alpha+\beta}(x) = (\alpha+\beta)x = \alpha \cdot x + \beta \cdot x = m_\alpha(x) + m_\beta(x)$$

and

$$m_\alpha \circ m_\beta(x) = m_\alpha(\beta \cdot x) = (\alpha\beta)x = m_{\alpha\beta}(x)$$

Since x is an arbitrary element of L we have $m_{\alpha+\beta} = m_\alpha + m_\beta$ and $m_{\alpha\beta} = m_\alpha \circ m_\beta$. Easily m_1 is the identity mapping since $1 \cdot \alpha = \alpha$ and therefore $m_1 \circ m_\alpha = m_\alpha$. ■

Theorem 1.3.2

The trace $\mathrm{Tr}_{L/K} : L \to K$ is a K-linear and the norm $\mathrm{N}_{L/K} : L \to K$ is multiplicative. Moreover, $\mathrm{N}_{L/K}(L^*) \subseteq K^*$.

Proof: By Lemma 1.3.1 for any $\alpha, \beta \in L$ we have $m_{\alpha+\beta} = m_\alpha + m_\beta$ and $m_{\alpha\beta} = m_\alpha \circ m_\beta$. Therefore, if we look into the matrix representation of m_α and m_β then we get

$$[m_{\alpha+\beta}] = [m_\alpha] + [m_\beta], \qquad [m_{\alpha\beta}] = [m_\alpha \circ m_\beta] = [m_\alpha] \cdot [m_\beta]$$

So

$$\mathrm{Tr}_{L/K}(\alpha+\beta) = \mathrm{Tr}([m_{\alpha+\beta}]) = \mathrm{Tr}([m_\alpha]) + \mathrm{Tr}([m_\beta]) = \mathrm{Tr}_{L/K}(\alpha) + \mathrm{Tr}_{L/K}(\beta)$$

and

$$\mathrm{N}_{L/K}(\alpha\beta) = \det([m_{\alpha\beta}]) = \det([m_\alpha]) \cdot \det([m_\beta]) = \mathrm{N}_{L/K}(\alpha) \cdot \mathrm{N}_{L/K}(\beta)$$

So $\mathrm{Tr}_{L/K}$ is additive and $\mathrm{N}_{L/K}$ is multiplicative.

Now to show $\mathrm{Tr}_{L/K}$ is K-linear we need to show for any $c \in K$ and $\alpha \in L$ we have $\mathrm{Tr}_{L/K}(c \cdot \alpha) = c \cdot \mathrm{Tr}_{L/K}(\alpha)$ which is true since

$$\mathrm{Tr}_{L/K}(c \cdot \alpha) = \mathrm{Tr}([m_{c\alpha}]) = \mathrm{Tr}(c[m_\alpha]) = c \cdot \mathrm{Tr}([m_\alpha]) = c \cdot \mathrm{Tr}_{L/K}(\alpha)$$

Now $\mathrm{N}_{L/K}(1) = 1$. So for any non-zero $\alpha \in L^*$, $\mathrm{N}_{L/K}(\alpha) \cdot \mathrm{N}_{L/K}(\alpha^{-1}) = 1$ and hence $\mathrm{N}_{L/K}(\alpha) \neq 0$. Therefore, $\mathrm{N}_{L/K}(L^*) \subseteq K^*$. ■

Now the elements $\mathrm{Tr}_{L/K}(\alpha)$ and $\mathrm{N}_{L/K}(\alpha)$ can be expressed in terms of the coefficients or the roots of the minimal polynomial of α over K. To explain this we need another polynomial in $K[X]$ which is very similar to the characteristic polynomial of any square matrix.

Definition 1.3.2: Characteristic Polynomial of $\alpha \in L$

For $\alpha \in L$ where L/K is a finite extension the characteristic polynomial *of α relative to the extension L/K is the characteristic polynomial of the matrix representation $[m_\alpha]$:*

$$P_{\alpha,L/K}(X) = \det(X \cdot I_n - [m_\alpha]) \in K[X]$$

where $n = [L : K]$. Therefore, $\deg(P_{\alpha,L/K}) = [L : K]$.

Note that here we defined the characteristic polynomial to be of the form $\det(X \cot I_n - A)$ instead of $\det(A - X \cdot I_n)$ which we typically do in linear algebra. This is because we want our polynomials to monic.

If A is a square matrix then the trace of A and the determinant of A appears as signed coefficients in its characteristic polynomial.

$$\det(X \cdot I_n - A) = X. - \mathrm{Tr}(A)X^{n-1} + \cdots + (-1)^n \det(A)$$

Therefore

$$P_{\alpha,L/K}(X) = X^n - \mathrm{Tr}_{L/K}(\alpha)X^{n-1} + \cdots + (-1)^n \mathrm{N}_{L/K}(\alpha) \tag{1.1}$$

Below we show a Cayley-Hamilton like theorem for characteristic polynomial of field elements.

Theorem 1.3.3

For every $\alpha \in L$, $P_{\alpha,L/K}(\alpha) = 0$.

Proof: By the consequence of Cayley-Hamilton theorem we have $P_{\alpha,L/K}([m_\alpha]) = O$. Now for any polynomial $f \in K[X]$ where $f(X) = X^n + \sum_{i=0}^{n-1} a_i X^i$ then

$$f([m_\alpha]) = [m_\alpha]^n + \sum_{i=0}^{n-1} a_i [m_\alpha]^i = [m_{\alpha^n}] + \sum_{i=0}^{n-1} [m_{a_i \alpha^i}] = m_{\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + c_0} = m_{f(\alpha)}$$

Therefore $P_{\alpha,L/K}([m_\alpha]) = m_{P_{\alpha,L/K}(\alpha)}$. Since $P_{\alpha,L/K}([m_\alpha]) = O$ and by Lemma 1.3.1 all m_α maps are distinct we have $P_{\alpha,L/K}(\alpha) = 0$. ■

Now for any $\alpha \in L$ let $f = \min(K, \alpha)$. Let $\deg(f) = d$. Hence, $[K(\alpha) \colon K] = d$. Therefore, $P_{\alpha,K(\alpha)/K}$ is a monic degree d polynomial over K. Since both f and $P_{\alpha,K(\alpha)/K}$ are monic and have same degree and shares a common root we have $f = P_{\alpha,K(\alpha)/K}$. So we have the following theorem.

Theorem 1.3.4

For any $\alpha \in L$, where L/K is a finite extension. Let $f = \min(K, \alpha)$ is the minimal polynomial of α in $K[X]$. If $L = K(\alpha)$ then $P_{\alpha,L/K}(X) = f(X)$.

More generally, $P_{\alpha,L/K}(X) = f^{n/d}$ where $d = \deg(f) = [K(\alpha) \colon K]$.

Proof: We already prove the first part in the discussion before the theorem. So we will show the second part here. Let $[L \colon K(\alpha)] = m$ and $\{\beta_1, \ldots, \beta_m\}$ is a basis of L over $K(\alpha)$. $1, \alpha, \ldots, \alpha^{d-1}$ is a basis of $K(\alpha)$ over K. Therefore, $\{\alpha^i \beta_j \colon 0 \le i \le d-1, j \in [n]\}$ is a basis of L over K. So for all $i \in \{0, 1, \ldots, d-1\}$ there exists $c_{i,j} \in \mathbb{F}_q$ such that $\alpha \cdot \alpha^i = \sum_{j=0}^{d-1} c_{ij}\alpha^j$. Therefore for any $t \in [n]$ and $i \in \{0, 1, \ldots, d-1\}$, $\alpha \cdot \alpha^i \beta_t = \sum_{j=0}^{d-1} c_{ij}\alpha^j \cdot \beta_t$. Hence, the matrix $[m_\alpha]$ is a repeated diagonal $d \times d$ block matrix of m_α for $K(\alpha)$ over K created by c_{ij}. Therefore, $P_{\alpha,L/K} = (P_{\alpha,K(\alpha)/K})^{n/d}$. ■

Corollary 1.3.5

Let the minimal polynomial of α over K be $X^d + c_{d-1}X^{d-1} + \cdots c_1 X + c_0$. Then

$$\mathrm{Tr}_{K(\alpha)/K}(\alpha) = -c_{d-1} \quad \textit{and} \quad \mathrm{N}_{K(\alpha)/k}(\alpha) = (-1)^d c_0$$

and more generally when $K \subseteq K(\alpha) \subseteq L$ and $[L \colon K] = n$,

$$\mathrm{Tr}_{L/K}(\alpha) = -\frac{n}{d} c_{d-1} = -[L \colon K(\alpha)] \cdot c_{d-1}, \quad \textit{and} \quad \mathrm{N}_{L/K}(\alpha) = (-1)^n \cdot c_0^{n/d} = (-1)^n \cdot c_0^{[L \colon K(\alpha)]}$$

Proof: Let f is the minimal polynomial of α over K. Hence by Theorem 1.3.4, $P_{\alpha,K(\alpha)/K} = f(X)$. So we have the first part. Now

$$P_{\alpha,L/K}(X) = f^{n/d}(X) = \left(X^d + c_{d-1}X^{d-1} + \cdots + c_1X + c_0\right)^{n/d} = X^n + \frac{n}{d}c_{d-1}X^{n-1} + \cdots + c_o^{n/d}$$

Therefore by comparing with (1.1) we have the result. ■

Observation 1.6. *We can rewrite the above formulas for* $\mathrm{Tr}_{L/K}(\alpha)$ *and* $\mathrm{N}_{L/K}(\alpha)$ *in terms of* $\mathrm{Tr}_{K(\alpha)/K}$ *and* $\mathrm{N}_{K(\alpha)/K}$ *in the following way*

$$\mathrm{Tr}_{L/K}(\alpha) = [L\colon K(\alpha)]\,\mathrm{Tr}_{K(\alpha)/K}(\alpha), \qquad \mathrm{N}_{L/K}(\alpha) = \mathrm{N}_{K(\alpha)\colon K}(\alpha)^{[L\colon K(\alpha)]}$$

From Corollary 1.3.5 we can draw a relation between the coefficients of the trace and the norm with the roots of the minimum polynomial of α. For any $\alpha \in L$ if f is the minimum polynomial of α over K then $P_{\alpha,K(\alpha)/K} = f(X)$. Now let $f(X) = X^d + c_{d-1}X^{d-1} + \cdots c_1X + c_0$ and f splits into linear factors $f(X) = (X-\alpha_1)\cdots(X-\alpha_d)$ in the splitting field of f over K. Then $-c_{d-1} = \alpha_1 + \cdots + \alpha_d$ and $(-1)^d c_0 = \alpha_1 \cdots \alpha_d$. Therefore, by rewriting Corollary 1.3.5 we get the following corollary.

Corollary 1.3.6

Let the minimal polynomial for α over K splits completely over a large enough field extension L of K as $(X-\alpha_1)\cdots(X-\alpha_d)$. Then

$$\mathrm{Tr}_{K(\alpha)/K}(\alpha) = \alpha_1 + \cdots + \alpha_d \quad \textit{and} \quad \mathrm{N}_{K(\alpha)/K}(\alpha) = \alpha_1 \cdots \alpha_d$$

More generally, if $K \subseteq K(\alpha) \subseteq L$, then

$$\mathrm{Tr}_{L/K}(\alpha) = \frac{n}{d}(\alpha_1 + \cdots + \alpha_d) \quad \textit{and} \quad \mathrm{N}_{L/K}(\alpha) = (\alpha_1 \cdot \alpha_d)^{n/d}$$

where $[L\colon K] = n$ and $[K(\alpha)\colon K] = d$, so $n/d = [L\colon K(\alpha)]$.

Now for finite fields for any $\alpha \in \mathbb{F}_{q^d}$ let f is the minimal polynomial of α over $\mathbb{F}_q$. Then by Theorem 1.2.4 we know the roots of f are $\alpha, \alpha^q, \ldots, \alpha^{\deg(f)-1}$. Hence, rewriting Corollary 1.3.6 in terms of conjugates of α we get the following corollary.

Corollary 1.3.7 Formula of Trace and Norm

Let $K/\mathbb{F}_q$ be finite extension with $[K\colon \mathbb{F}_q] = n$. Then for any $\alpha \in K$,

$$\mathrm{Tr}_{K/\mathbb{F}_q} = \alpha + \alpha^q + \cdots + \alpha^{q^{n-1}} \quad \textit{and} \quad \mathrm{N}_{K/\mathbb{F}_q}(\alpha) = \alpha \cdot \alpha^q \cdots \alpha^{q^{n-1}}$$

In particular, the trace and norm are invariant under the Frobenius automorphism $\alpha \mapsto \alpha^q$: replacing α by α^q merely cycles the summands (respectively, the factors) in the expressions above, leaving the total unchanged.

Theorem 1.3.8

Let K be a finite extension of $F = \mathbb{F}_q$. The F-linear transformations from K into F are exactly the mappings L_β, $\beta \in K$, where $L_\beta(\alpha) = \mathrm{Tr}_{K/F}(\beta\alpha)$ for all $\alpha \in K$. Furthermore, $L_\beta \neq L_\gamma$ whenever β and γ are distinct elements of K.

Proof: For any $\alpha_1, \alpha_2 \in K$ and $c \in F$,

$$L_\beta(\alpha_1 + c\alpha_2) = \mathrm{Tr}_{K/F}(\beta(\alpha_1 + c\alpha_2)) = \mathrm{Tr}_{K/F}(\beta\alpha_1) + c\ \mathrm{Tr}_{K/F}(\beta\alpha_2) = L_\beta(\alpha_1) + c\,L_\beta(\alpha_2),$$

using the F-linearity of $\mathrm{Tr}_{K/F}$ from Theorem 1.3.2.

Let $\beta, \gamma \in K$ with $\beta \neq \gamma$. For any $\alpha \in K$,

$$L_\beta(\alpha) - L_\gamma(\alpha) = \mathrm{Tr}_{K/F}(\beta\alpha) - \mathrm{Tr}_{K/F}(\gamma\alpha) = \mathrm{Tr}_{K/F}((\beta-\gamma)\alpha).$$

Since $\beta - \gamma \neq 0$, the map $\alpha \mapsto (\beta-\gamma)\alpha$ is a K-linear bijection on F, so the image of $\alpha \mapsto \mathrm{Tr}_{F/K}((\beta-\gamma)\alpha)$ equals the image of $\mathrm{Tr}_{F/K}$ itself. As $\mathrm{Tr}_{F/K} : F \to K$ is a non-zero K-linear map onto K, since $\mathrm{Tr}_{K/F}(1) = [K\colon F] > 0$, there exists $\alpha \in F$ such that $\mathrm{Tr}_{F/K}((\beta-\gamma)\alpha) \neq 0$. Hence, $L_\beta \neq L_\gamma$.

Write $F = \mathbb{F}_q$ and $K = \mathbb{F}_{q^m}$, so $[K\colon F] = m$. Since the L_β are pairwise distinct as β ranges over $K = \mathbb{F}_{q^m}$, they yield q^m distinct F-linear maps from K to F. On the other hand, any F-linear map $K \to F$ is determined by its values on a fixed F-basis of K, which has m elements; each value can be assigned arbitrarily in $F = \mathbb{F}_q$. So the total number of F-linear maps $K \to F$ is q^m. Therefore, the maps $\{L_\beta : \beta \in K\}$ exhaust all F-linear transformations from K into F.

■

Theorem 1.3.9

Let K be a finite extension of $F = \mathbb{F}_q$. Then for $\alpha \in K$ we have $\mathrm{Tr}_{K/F}(\alpha) = 0$ if and only if $\alpha = \beta^q - \beta$ for some $\beta \in K$.

Proof: The sufficiency of the condition is obvious. To prove the necessity, suppose $\alpha \in K = \mathbb{F}_{q^m}$ with $\mathrm{Tr}_{K/F}(\alpha) = 0$ and let β be a root of $X^q - X - \alpha$ in some extension field of K. Then $\beta^q - \beta = \alpha$ and

$$\begin{aligned} 0 = \mathrm{Tr}_{K/F}(\alpha) &= \alpha + \alpha^q + \cdots + \alpha^{q^{m-1}} \\ &= (\beta^q - \beta) + (\beta^q - \beta)^q + \cdots + (\beta^q - \beta)^{q^{m-1}} \\ &= (\beta^q - \beta) + (\beta^{q^2} - \beta^q) + \cdots + (\beta^{q^m} - \beta^{q^{m-1}}) \\ &= \beta^{q^m} - \beta, \end{aligned}$$

so that $\beta \in K$. ■

Theorem 1.3.10

Let $L/K/F$ be finite extensions. Then for any $\alpha \in L$

$$\mathrm{Tr}_{L/F}(\alpha) = \mathrm{Tr}_{K/F} \circ \mathrm{Tr}_{L/K}(\alpha) \quad \textit{and} \quad \mathrm{N}_{L/F}(\alpha) = \mathrm{N}_{K/F} \circ \mathrm{N}_{L/K}(\alpha)$$

Proof: Let $[L\colon K] = m$ and $[K\colon F] = n$. Suppose $\{a_1, \ldots, a_n\}$ is a F-basis of K and $\{b_1, \ldots, b_m\}$ is a K-basis of L. Then $\{a_i \cdot b_j \colon i \in [n], j \in [m]\}$ is a F-basis of L.

For any $\alpha \in L$ let

$$\alpha \cdot b_j = \sum_{i=1}^{m} c_{ij} \cdot b_i, \quad c_{ij} a_s = \sum_{r=1}^{n} d_{ijrs} \cdot a_r \tag{1.2}$$

where $c_{ij} \in K$ and $d_{ijrs} \in F$. Therefore,

$$\alpha(a_s b_j) = \sum_{i=1}^{m} c_{ij} \cdot a_s \cdot b_i = \sum_{i=1}^{m} \sum_{r=1}^{n} b_{ijrs} \cdot a_r b_j$$

So now if we look into the matrix representation of $[m_\alpha]_{L/K}$ and $[m_\alpha]_{L/F}$ then we have

$$[m_\alpha]_{L/K} = (c_{ij})_{1 \le i,j \le n}, \quad [m_{c_{ij}}]_{K/F} = (b_{ijrs})_{1 \le r,s \le n}, \quad [m_\alpha]_{L/F} = \left([m_{c_{ij}}]_{K/F}\right)_{1 \le i,j \le n}$$

where the field extension in the subscript indicates what extension is being used for that matrix. The last matrix is a block matrix. So using all these three matrices we get

$$\mathrm{Tr}_{K/F} \circ \mathrm{Tr}_{L/K}(\alpha) = \mathrm{Tr}_{K/F}\left(\sum_{i=1}^{m} c_{ii}\right) = \sum_{i=1}^{m} \mathrm{Tr}_{K/F}(c_{ii}) = \sum_{i=1}^{m} \mathrm{Tr}\left([m_{c_{ii}}]_{K/F}\right) = \sum_{i=1}^{m}\sum_{r=1}^{n} b_{iirr} = \mathrm{Tr}_{L/F}(\alpha)$$

So we have the transitivity of the trace function. Now we will show for the norm function. For norm it is very delicate and harder to show. We will show a proof from [Sch05]. Again assume $[L\colon K] = m$ and $[K\colon F] = n$. We will prove this by dividing it into 3 cases.

Case I $\alpha \in K$**:** Suppose the transitivity formula already holds. Then $\mathrm{N}_{L/K}(\alpha) = \alpha^{[L\colon K]}$ and hence $\mathrm{N}_{K/F} \circ \mathrm{N}_{L/K}(\alpha) = \mathrm{N}_{K/F}(\alpha^{[L\colon K]}) = \mathrm{N}_{K/F}(\alpha)^{[L\colon K]}$. So in this case we will try to show that this expression is true.

In this case $F \subseteq F(\alpha) \subseteq K \subseteq L$. Let $f - \min(F, \alpha)$ and $\deg(f) = d$. Therefore, by Theorem 1.3.4 we have $P_{\alpha,F(\alpha)/F}(X) = f(X)$ and $P_{\alpha,K/F}(X) = P_{\alpha,F(\alpha)/F}(X)^{[K\colon F(\alpha)]}$ and $P_{\alpha,L/F}(X) = P_{\alpha,F(\alpha)/F}(X)^{[L\colon F(\alpha)]}$. Since $F \subseteq F(\alpha) \subseteq K \subseteq L$ we have by Theorem 1.1.6, $[L\colon F(\alpha)] = [L\colon K] \cdot [K\colon F(\alpha)]$. Therefore,

$$P_{\alpha,L/F}(X) = P_{\alpha,F(\alpha)/F}(X)^{[L\colon F(\alpha)]} = \left(P_{\alpha,F(\alpha)/F}(X)^{[K\colon F(\alpha)]}\right)^{[L\colon K]} = P_{\alpha,K/F}(X)^{[L\colon K]}$$

So if we compare the constant terms we get

$$(-1)^{[L\colon F]}\,\mathrm{N}_{L/F}(\alpha) = \left((-1)^{[K\colon F]}\,\mathrm{N}_{K/F}(\alpha)\right)^{[L\colon K]} \implies \mathrm{N}_{L/F}(\alpha) = \mathrm{N}_{K/F}(\alpha)^{[L\colon K]}$$

Hence we have settled this case.

Case II $L = K(\alpha)$**:** Let f is the minimal polynomial of α over K. Then $\deg(f) = m$. Let $f(X) = X^m + c_{m-1}X^{m-1} + \cdots + c_0$. Then

$$\mathrm{N}_{K/F} \circ \mathrm{N}_{K(\alpha)/K}(\alpha) = \mathrm{N}_{K/F}((-1)^m c_0) = (-1)^{mn}\,\mathrm{N}_{K/F}(c_0)$$

So now we will compute $\mathrm{N}_{K/F}(c_0)$.

Let $\{b_1, \ldots, b_n\}$ is an F-basis of K. So an F-basis of $K(\alpha)$ is $\{\alpha^i \cdot b_j\colon 0 \le i \le m-1, 1 \le j \le n\}$. By definition $\mathrm{N}_{K(\alpha)/F}(\alpha)$ is the determinant of the matrix for multiplication by α on $K(\alpha)$ as an F-linear map. Now for all $j \in [n]$, $\alpha \cdot (\alpha^i b_j) = \alpha^{i+1} b_j$ where $0 \le i \le m-2$ but

$$\alpha \cdot (\alpha^{m-1} b_j) = \alpha^m b_j = -\alpha^{m-1}(c_{m-1} b_j) - \cdots - \alpha(c_1 b_j) - c_0 b_j$$

Let C_i be the matrix of expressing $c_i b_j$ for all $j \in [n]$ as F-linear combinations of $b_1, \ldots, b_n$. Hence, using the above basis we have the matrix to be

$$[m_\alpha]_{K(\alpha)/F} = \begin{bmatrix} O & O & \cdots & O & -C_0 \\ I_n & O & \cdots & O & -C_1 \\ O & I_n & \cdots & O & -C_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ O & O & \cdots & I_n & -C_{m=1} \end{bmatrix} = \begin{bmatrix} O & -C_0 \\ I_{(m-1)n} & B \end{bmatrix}$$

where O is the zero matrix of dimension $n \times (m-1)n$ and B is the $(m-1)n \times n$ matrix gathering all the $C_1, \ldots, C_{m-1}$. Therefore,

$$\det \begin{bmatrix} O & -C_0 \\ I_{(m-1)n} & B \end{bmatrix} = (-1)^{(m-1)n^2} \det(-C_0) = (-1)^{(m-1)n} \cdot (-1)^n \det(C_0) = (-1)^{mn} \det(C_0)$$

Now C_0 is the matrix for multiplication by c_0 and therefore, $\det(c_0) = \mathrm{N}_{K/F}(c_0)$. So, $\mathrm{N}_{K(\alpha)/F}(\alpha) = (-1)^{mn}\,\mathrm{N}_{K/F}(c_0)$ and previously we prove that $(-1)^{mn}\,\mathrm{N}_{K/F}(c_0) = \mathrm{N}_{K/F} \circ \mathrm{N}_{K(\alpha)/K}(\alpha)$. So we have settled case II.

Case III General Situation: Therefore in this case $F \subseteq K \subseteq F(\alpha) \subseteq L$. Then we have

$$\begin{aligned}
\mathrm{N}_{L/F}(\alpha) &= \mathrm{N}_{F(\alpha)/F}(\alpha)^{[L:\,F(\alpha)]} && \text{[By Case I for } L/F(\alpha)/F] \\
&= \left(\mathrm{N}_{K/F} \circ \mathrm{N}_{F(\alpha)/K}(\alpha)\right)^{[L:\,F(\alpha)]} && \text{[By Case II for } F(\alpha)/K/F] \\
&= \mathrm{N}_{K/F}\left(\mathrm{N}_{F(\alpha)/K}(\alpha)^{[L:\,F(\alpha)]}\right) \\
&= \mathrm{N}_{K/F} \circ \mathrm{N}_{L/K}(\alpha) && \text{[By Case I for } L/F(\alpha)/F]
\end{aligned}$$

This settles case III too. And hence the proof is complete. ■

This concludes our discussion on finite fields and their basic theory, including field extensions, the structure of $\mathbb{F}_q$, and the trace and norm maps. In the next chapter we turn to the theory of characters of finite fields, which will be the primary tool in our study of equations over finite fields. For a thorough and self-contained treatment of finite fields beyond what is covered here, the reader is referred to [LN96].

Chapter 2

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